



Choupo

The Open Source Chemical Process Simulator

Properties Guide

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“Thermodynamics is a funny subject. The first time you go through it, you don’t understand it at all. The second time you go through it, you think you understand it, except for one or two points. The third time you go through it, you know you don’t understand it, but by that time you are so used to it, it doesn’t bother you any more.”

— Arnold Sommerfeld

1 Introduction: the difficulty, the map, the promise

What this guide is — and is not. This is the props-**mode workflow** guide: how you *estimate* a missing compound, *fit* its binary pairs, and *consistency-test* the data — the see-then-decide loop of the Props/Explore workspace. It is *not* the property *theory*. For the first-principles derivations of NRTL, Wilson, UNIQUAC, UNIFAC, the cubic equations of state (SRK, Peng–Robinson), the electrolyte models (Pitzer, eNRTL, aqueous speciation), Henry’s law, Rackett, Ambrose–Walton, sublimation and the NASA–CEA Gibbs-of-formation data, see the **Theory Guide** (the in-GUI model panels deep-link straight into it).

If thermodynamics felt slippery the first time — the formalism pristine, yet somehow never self-sufficient when you sat down with real data — you are in good company: Sommerfeld, who reshaped quantum physics, said he never wrote his thermodynamics book because he never felt he had understood the subject. The difficulty is real, but it is not formal incompleteness. It is that an *axiomatic-looking* macroscopic theory rests, underneath, on emergent statistical facts and on parameters that must be *measured*, not derived. The confusion is the seam between the two showing through. This guide does not hide that seam — it makes it the organising idea.

1.1 The target: where theory grips

Picture property prediction as a target (Figure 1): pristine at the core, increasingly empirical toward the rim.



Figure 1: The layers of property prediction. The two inner shells follow from the laws alone. The accent shell — *constitutive models* — is the first fault line: the laws say *that* fugacities must match across phases, never *how* to compute one for a real mixture. From there outward, the theory is necessary but no longer sufficient.

- **The four laws (core).** The zeroth, first, second and third laws define temperature, internal energy, entropy, and — via Nernst — an absolute zero of entropy. Parameter-free.
- **Formalism.** Pure manipulation yields the potentials H , G , A , the Maxwell relations, the equilibrium criteria (equality of fugacity across phases, $\Delta G = -RT \ln K$ for reaction) and the phase rule. Still derivable from nothing but the laws.
- **Constitutive models — where it first grips.** The laws require fugacities to be equal across phases; they do not tell you how to *compute* a fugacity for a real mixture. You must supply a model they do not give you — an equation of state, an activity model, a kinetic law. In Choupo: `idealGas/SRK/PR`; `ideal/NRTL/Wilson/UNIFAC`.
- **Data — “go to the laboratory.”** Those models carry parameters thermodynamics cannot predict: T_c , P_c , ω , the Antoine constants, $C_p^\circ(T)$, ΔH_f° , S° , the NRTL a_{ij} . Each is measured or estimated, and each has a finite

validity range. In Choupo, these are the `.dat` files.

- **Process & numerics.** Where the predicted numbers are consumed — flash, distillation, recycle — and grip a second time, now numerically (multiple EoS roots, phase stability, initialisation).

Key idea. Sommerfeld’s difficulty is this picture: a theory that *looks* axiomatic to the rim, but stops being self-sufficient at the accent shell. The honest move is not to paper over the seam between rigorous derivation and fitted correlation — it is to show exactly where it runs.

1.2 The glass-box promise

Because the outer shells are empirical, every number Choupo predicts should travel with two things attached:

- **Provenance** — which method produced it (rigorous derivation, fitted correlation, or group-contribution estimate) and the source of each parameter (a measured datum or an estimate).
- **Validity awareness** — whether a correlation is *extrapolating beyond its fitted range*, with a warning when it is, rather than a confident number where none is warranted.

Intuition. Ask Choupo for the vapour pressure of nitrogen at room temperature and it *refuses*: N_2 is supercritical there (its critical temperature is -147°C), and a saturation pressure does not exist above the critical point. The P^{sat} curve simply *breaks* at T_c instead of extrapolating Antoine into a meaningless ~ 600 bar. That refusal is the whole promise in miniature: the model declines to invent a number the physics forbids.

1.3 Where we are heading

A single property over a wide temperature span is not one problem — it is several physical regimes stitched together, and your confidence in any predicted number rises and collapses as you cross them. Figure 2 is the destination of this guide: the vapour pressure of a **water / ethanol / benzene** mixture from -200 to 1000°C , drawn not as a curve but as a band of *confidence*.

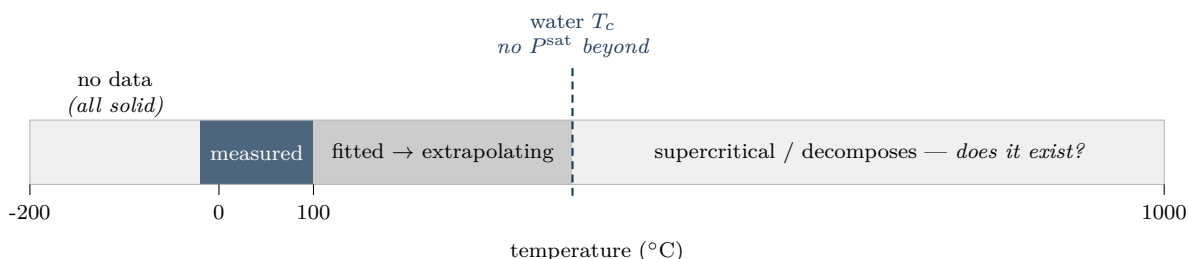


Figure 2: The same property, five honesties. A narrow window where we actually have data; degradation outward as the correlation extrapolates; a hard *wall* at the critical temperature where the property ceases to exist (the nitrogen refusal, generalised); and a cryogenic desert where every species has frozen solid (ethanol last, at -114°C) and there is simply no data. The theory runs to both edges; reality and data do not.

We build each piece of that picture in the chapters that follow — estimating a missing component (§6), curating the mixture models against lab points (§7), and testing whether the data itself is even consistent (§8). We read that span regime by regime at the end of this introduction (§1.6), and return to solve it numerically, end to end, at the close.

1.4 Anatomy of a component: where the numbers come from

Open any `.dat` and you see a flat list of numbers. The natural question — the one this whole guide turns on — is: *which of these are independent inputs, and which derived properties does the engine actually build from them?* Let us read `acetone.dat` field by field and follow each number to the property it feeds and to the line of code that consumes it.

Field	What it is, and what it builds
<i>Identity</i>	
MW	molar mass; converts molar $\leftrightarrow$ mass everywhere.
<i>Caloric anchors — standardThermochemistry</i>	
dHf_298 ΔH_f°	the <i>position</i> of the enthalpy scale: bond energy C_p knows nothing about (combustion calorimetry).
s_298 S°	the <i>absolute</i> (third-law) entropy at 298 K — a true zero-referenced value, not a difference.
<i>Caloric shape — idealGasHeatCapacity</i>	
coefficients $C_p^\circ(T)$	the master correlation: integrated to give the <i>temperature shape</i> of H , S , G (below).
<i>Corresponding-states trio</i>	
Tc, Pc	critical point: set the <i>size</i> of the cubic-EoS attraction term a_c , and the reduced temperature $T_r = T/T_c$.
omega ω	acentric factor: sets the <i>temperature shape</i> of that attraction, $\alpha(T_r; \omega)$ — how non-spherical / polar the molecule is.
<i>Phase change</i>	
Tb, HvapTb	boiling point + latent heat at T_b ; with Tc drive Watson's $\Delta H_{\text{vap}}(T)$.
Vliq	liquid molar volume (Poynting, liquid density).
<i>Correlations & recipes</i>	
vaporPressure	Antoine $\log_{10} P^{\text{sat}} = A - B/(T + C)$, with its Trange ; the saturation curve.
groups	the molecular decomposition a group-contribution estimator / UNIFAC reads (the curation recipe, §6).
diffusionVolume	Fuller gas diffusivity $\Rightarrow$ Lewis number (wet-bulb).

The caloric engine: H , S , G from C_p

This is the mechanism behind “other properties that derive from here” — and it is worth seeing *how one arrives at* $S(T)$, since that is the step that trips everyone up. Start from the laws, at constant pressure. Heat capacity *is* the temperature slope of enthalpy, $C_p = (\partial H / \partial T)_P$, so

$$dH = C_p dT.$$

For entropy, the second law gives $dS = \delta q_{\text{rev}}/T$; at constant pressure the reversible heat is $\delta q_{\text{rev}} = C_p dT$, hence

$$dS = \frac{C_p}{T} dT.$$

That single step — dividing by T — is the whole origin of the $\int C_p/T$. Integrate each from a reference temperature and add the 298 K anchor; Choupo stores $C_p^\circ(T)$ and the two anchors (*not* tables of H , S) and evaluates:

$$H^\circ(T) = \underbrace{\Delta H_f^\circ}_{\text{position}} + \int_{298.15}^T C_p^\circ(T') dT' \quad (\text{Component}::\text{h_pure_ig}, \text{Component.cpp}) \quad (1)$$

$$S^\circ(T) = \underbrace{S_{298}^\circ}_{\text{absolute}} + \int_{298.15}^T \frac{C_p^\circ(T')}{T'} dT' \quad (\text{Component}::\text{s_pure_ig}) \quad (2)$$

$$G^\circ(T) = H^\circ(T) - T S^\circ(T) \quad (3)$$

For a mixture the engine adds the entropy of mixing $-R \sum y_i \ln y_i$ and the pressure term $-R \ln(P/P_{\text{ref}})$ (**ThermoPackage::S_ig**). That is the *entire* ideal-gas caloric model — no hidden tables.

These integrals are not done numerically. For a polynomial C_p the primitive is elementary — it was solved *once, on paper* — so each call evaluates a closed form,

$$\int_{T_{\text{ref}}}^T \frac{C_p^\circ}{T'} dT' = a_0 \ln \frac{T}{T_{\text{ref}}} + \sum_{k \geq 1} \frac{a_k}{k} (T^k - T_{\text{ref}}^k),$$

exact to machine precision in a handful of operations (**PolynomialCp::S**). The standard correlations all share this property (NASA, Aly–Lee, Shomate have closed primitives too); Choupo never runs a quadrature loop for C_p .

Worked example 1.1. Acetone H and S at 400 K — by hand, then by ChoupoWith $C_p^\circ(T) = 6.301 + 0.2606 T - 1.253 \times 10^{-4} T^2 + 2.038 \times 10^{-8} T^3$ (units $\text{J mol}^{-1} \text{K}^{-1}$), integrating from 298.15 to 400 K:

$$\int C_p^\circ dT' = 8431 \text{ J/mol}, \quad \int \frac{C_p^\circ}{T'} dT' = 24.19 \text{ J/(mol} \cdot \text{K)}.$$

So $H^\circ(400) = -217100 + 8431 = -208669$ J/mol and $S^\circ(400) = 295.30 + 24.19 = 319.49$ J/(mol · K). Running `choupoProps` on a one-point scan returns exactly $H_{\text{ig}} = -208669$ J/mol, $S_{\text{ig}} = 319.49$ J/(mol · K). The number on the page is the number in the code — and at 298.15 K both integrals vanish, so $H = \Delta H_f^\circ$ and $S = S_{298}^\circ$, the anchors laid bare.

Key idea. C_p gives the *shape*; the formation values give the *position*. That is why the `.dat` lists $C_p^\circ(T)$ and ΔH_f° , S° separately — not redundancy. And note the asymmetry: entropy has an absolute zero for free (third law: $S \rightarrow 0$ for a perfect crystal at $T=0$), so S_{298}° is a genuine absolute; enthalpy has *no* natural zero, so ΔH_f° must be *measured* and supplied.

What the critical properties and acentric factor actually do

T_c , P_c , ω do nothing in the caloric equations above — they enter only when you ask for *real-fluid* behaviour through a cubic equation of state. There the attraction term is $a(T) = a_c \alpha(T_r)$, with a_c fixed by T_c , P_c and the temperature shape fixed by ω (Soave/SRK):

$$\alpha(T) = \left[1 + m(1 - \sqrt{T_r}) \right]^2, \quad m = 0.48508 + 1.55171\omega - 0.15613\omega^2, \quad T_r = \frac{T}{T_c}$$

(`SRK.cpp`; Peng–Robinson uses the analogous $\kappa(\omega)$). This $a(T)$ is what produces the departure functions $H - H^\circ$, $S - S^\circ$ and the compressibility Z — the real-gas correction layered on top of the ideal-gas caloric result. The same T_c, T_b also drive Watson’s latent heat, $\Delta H_{\text{vap}}(T) = \Delta H_{\text{vap}}(T_b) [(T_c - T)/(T_c - T_b)]^{0.38}$ (`Component.cpp`).

Intuition. So why does `acetone.dat` carry ω at all if you are running the default `idealGas`? Because the data is *model-agnostic* and the model decides which fields wake up. Under `idealGas` there is no departure, and ω, T_c, P_c are simply never read. Select `SRK` for a high-pressure compressor and the very same numbers come alive. The `.dat` is the laboratory record; the model is the question you ask of it.

1.5 From G to equilibrium: what ΔG° is (and is not)

The single most common confusion in this whole subject is ΔG° versus ΔG . It is worth settling once, because G is what the reactors minimise and it is built from exactly the H and S above.

First, the $^\circ$. It means *reference state* — each species pure, ideal gas, at 1 bar — but *at the reaction temperature* T . It does *not* mean “at 298 K.” So $\Delta G^\circ = \Delta G^\circ(T)$ varies with temperature; $\Delta G^\circ(298)$ is just one point on a curve.

Then, the two are not the same thing.

- ΔG° is the Gibbs change if reactants $\rightarrow$ products with *everything in its standard state*. It is a *fixed number* — a property of the reaction at T .
- ΔG is the *actual* Gibbs change at the real composition (the real partial pressures). It is the *driving force*.

The reaction quotient Q bridges them, and equilibrium is where the driving force — not the reference number — vanishes:

$$\Delta G = \Delta G^\circ + RT \ln Q, \quad \Delta G = 0 \text{ at equilibrium} \Rightarrow \boxed{\Delta G^\circ = -RT \ln K}$$

Key idea. At equilibrium it is ΔG that is zero, *never* ΔG° . ΔG° stays fixed and merely sets *where* the equilibrium sits (the value of K); the system gets there by shifting Q until $\Delta G = 0$. And $\Delta G^\circ > 0$ does not mean “no reaction” — it means $K < 1$, i.e. partial conversion. Only the sign and size of K change.

Where the number comes from — straight back to the caloric engine:

$$\Delta G^\circ(T) = \Delta H^\circ(T) - T \Delta S^\circ(T) = \sum_i \nu_i g_i^\circ(T), \quad g_i^\circ(T) = h_i^{\text{ig}}(T) - T s_i^{\text{ig}}(T).$$

Note the subtlety that confuses everyone: h^{ig} carries the *formation* enthalpy (ΔH_f° , an elements datum) while s^{ig} is the *absolute* (third-law) entropy. Mixing the two references looks wrong — but it is exactly right, because in a *balanced* reaction the element contributions cancel through $\sum_i \nu_i$. That is precisely why the `.dat` carries ΔH_f° and S° separately: each plays a different role, and they combine correctly only across a balanced stoichiometry (`Component::g_pure_ig`; consumed in `PFR.cpp`/`CSTR.cpp` as $K = \exp(-\Delta G^\circ/RT)$ and in the Gibbs reactor’s direct G -minimisation).

How K moves with T (Gibbs–Helmholtz). $\Delta G^\circ(T)$ is not a frozen datum — it has its own temperature law, and that law is a workhorse relation worth knowing by name. Gibbs–Helmholtz states $[\partial(G/T)/\partial T]_P = -H/T^2$; applied to $\Delta G^\circ = -RT \ln K$ it collapses to the van 't Hoff equation,

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

which is just Le Chatelier made quantitative: an exothermic reaction ($\Delta H^\circ < 0$) has K *falling* as T rises. In Choupo it doubles as a glass-box self-check — the $K(T)$ the reactors build from $g^\circ(T)$ at each temperature must show exactly this slope, or the enthalpy and the equilibrium constant are disagreeing with each other.

Intuition. These relations are not ornaments. Gibbs–Helmholtz is *how the equilibrium moves with temperature* (above); its sibling, the **Gibbs–Duhem** equation $\sum_i x_i d \ln \gamma_i = 0$ (constant T, P), says the activity coefficients of a mixture are *not independent*. That single constraint is exactly why measured VLE data can be *tested* for thermodynamic consistency — and why an activity model is built from one G^E function rather than fitting each γ_i freely. It is the engine of §8.

1.6 A worked span: water / ethanol / benzene, -200 to 1000°C

Put the three together and sweep the temperature. This one mixture crosses *every* regime of property prediction, and is the honest way to see both how properties are obtained *and* where the models give out — which is exactly the colour band of Figure 2, now read regime by regime. (Critical temperatures: ethanol 241°C , benzene 289°C , water 374°C ; melting points: ethanol -114 , water 0 , benzene 5.5°C .)

Regime	How a property is obtained	Where the model gives out
-200 to $\sim 0^\circ\text{C}$ solids (SLE)	melting points T_m , ΔH_{fus} , solid C_p ; freezing-point depression; sublimation P^{sat}	low-T edge. Below every Antoine Trange : P^{sat} would extrapolate blindly. Solid-phase data is sparse — a cryogenic <i>data desert</i> .
~ 0 to $\sim 65^\circ\text{C}$ two liquids (LLE)	activity model (NRTL/UNIFAC) for γ ; water/benzene nearly immiscible, ethanol the cosolvent	VLE-fitted parameters do <i>not</i> serve LLE; Wilson cannot represent a liquid split at all; without T -dependent parameters the miscibility gap appears/vanishes wrongly.
~ 65 to $\sim 100^\circ\text{C}$ boiling (VLLE)	Antoine P^{sat} + γ - ϕ + three-phase flash (Gibbs minimisation); ethanol/water + ternary azeotrope ($\sim 64.9^\circ\text{C}$)	azeotrope sensitivity — a small γ error moves the column design; bubble-point (Wang–Henke) distillation is unstable <i>through</i> an azeotrope (use method simultaneous).
~ 100 to $\sim 290^\circ\text{C}$ vapour $\rightarrow$ near-critical	cubic EoS (SRK/PR) with k_{ij} ; departure functions $H - H^\circ$, $S - S^\circ$, Z	cubics degrade as the critical region is approached; k_{ij} often needs to be made temperature-dependent over so wide a span.
~ 290 to 374°C mixed super/sub-critical	the method itself must switch γ - $\phi \rightarrow \phi$ - ϕ (one EoS for every phase)	above a species' T_c the activity-coefficient picture loses meaning; no single γ model spans the switch; near-critical accuracy is poor.
374 to 1000°C supercritical + re-action	ϕ - ϕ EoS; wide-range $C_p^\circ(T)$ for the caloric terms	high-T edge. P^{sat} ceases to exist above T_c (Choupo <i>refuses</i> it, §1.2); and the species thermally <i>decompose</i> — the fixed-component assumption fails: do they still exist?

The two edges — the ones you asked to watch — fail for opposite reasons but teach the same lesson. At the **cold edge** there is simply no data: every correlation is fitted to a window that starts well above 73 K , the species are frozen, and a confident number there would be invention, not prediction. At the **hot edge** the theory keeps producing numbers — Antoine will gladly report a vapour pressure at 800°C — but the *physics* has moved on: past each T_c no saturation exists, and past a few hundred degrees the molecules are no longer the molecules in the **.dat**. In both cases the correct output is a *flag*, not a figure.

Common pitfall. No single model spans this axis. A run that quietly uses one γ model and one cubic from -200 to 1000°C is not “general” — it is extrapolating through three regime changes in silence. The wide span is not one problem; it is five problems wearing one temperature axis.

Intuition. What Choupo ships *today* covers the middle of this span honestly: cubic EoS, ideal/NRTL/Wilson/UNIFAC activity, Antoine P^{sat} , Watson ΔH_{vap} , three-phase VLLE by Gibbs minimisation, and azeotropic columns via the simultaneous MESH solver. It does *not* (yet) carry association models

(PC-SAFT/CPA) for the hydrogen-bonding non-ideality, nor electrolyte or wide-range near-critical packages. So for the H-bonding mid-span and the two edges, Choupo's job is not to fake accuracy it does not have — it is to predict where it can and to *say so* where it cannot. That honesty is the product.

2 The Choupo property architecture — the complete map

Everything in this guide hangs off one architecture. This chapter is the complete map, stated once; every later chapter is a room in this house.

2.1 The flow

`constant/propertyPackage declares` → the builder *loads, verifies and announces* (it never estimates)
→ the engine *computes* → the run log *confesses*.

Estimation is a curation-time act (the props bench writes a reviewable `.dat`); assembly-time refusal is loud (a missing declared parameter file, a k_{ij} regressed for another cubic, a vapour method the liquid path cannot serve). The log then shows the whole assembly: the VLE world, the reference rung of every group, and every pair parameter with its source file.

2.2 The grammar — the axes of a property package

Axis	Slot	What it fixes
components	<code>components (...)</code>	identity; a salt's dissociations
phases model per PHASE	<code>phases, phases/solid data propertyMethods { liquid ...; vapour ...; }</code>	which phases exist; ρ_p, k_v for ONE Gibbs surface per phase slot selects the VLE world
convention per GROUP	reference rungs (<code>referenceBasis</code>)	renormalisations of one surface Raoult, solutes on Henry, ion infinite dilution
correlation per COMPONENT	the component <code>.dat</code> (+ overrides)	pure-species data; legal for pure correlations, never for the surface
parameter per PAIR	<code>parameters { ...Pairs {...} }</code>	declared FILE, verified at assembly to its source
mixing per PROPERTY × PHASE	<code>mixing slots + effective{}</code>	how pures combine (Wilke-Nissan, van-der-Waals one-fluid closures (a suspension's Thoma viscosity) live in <code>effective{}</code> are <i>media closures</i> , not thermodynamic
chemistry package per UNIT	<code>chemistry { salts (...); } unit thermo { }</code>	real equilibria with $K(T)$ and the spatial axis: a unit selection whole package — never a pair on single axes (that is the road to in one phase); the model build dited (ΔH at the seam)
phase equilibrium	<code>phaseEquilibrium { }</code>	for each coexisting phases imposed equality and the DATUM (e.g. <code>link datum aqueousInfDil; tra <pair-file>;</code>) — the bridge surfaces, first-class and default implicit

Provenance is an *attribute of every value*, not an axis: each resolved number carries a quality word (`measured > fitted > correspondingStates > estimated > asserted`), and an `asserted` value — a pinned constant, a user polynomial, a user table — is refused without its written *why*.

2.3 The four VLE worlds

All selected by the *liquid* method slot; each validated against measured data in this guide:

World	Liquid method	K-value	Natural home
γ - φ	activity.NRTL (Wilson, UNIQUAC, UNIFAC)	$\gamma_i P_i^{\text{sat}}/P$	organic mixtures
Henry	solution.henryDilute	$\gamma_i^* H_i/P$	dissolved gases
φ - φ	eos.SRK / eos.PengRobinson	φ_i^L/φ_i^V	high pressure, near-critical
electrolyte	electrolyte.pitzer / .eNRTL	ion activities, osmotic	brines, salts, mixed solvent

2.4 The five component routes

1. **Standard, by name** — `data/standards/components/` (exact-name $O(1)$, every value cited, the engine refuses to write there).
2. **Local, by name** — `data/local/components/` is the *private* working tier (gitignored; shipped empty). Records are runtime-usable, but every use is announced as `[local] UNVERIFIED`; a local record never shadows a standard one. Integrity and provenance gates make a local record inspectable, not automatically authoritative.
3. **New, defined in the case** — `myCase/constant/components/`: a new molecule (estimated and labelled), a new ion, a pseudo-component lump. The case runs self-contained from day one; promotion into the standards tree is a *requested, reviewed curation act* after validation.
4. **Standard + overlay** — a partial case file merges field-by-field over the standard entry (sample-specific measured data, a new model block); the run announces the merge.
5. **Solids** — solid reference rungs, solids on their own stream channel, and the iron rule: a dissolving salt's solid formation enthalpy is ION-DERIVED ($H_f^{\text{solid}} = \sum \nu_i h_{f,aq,i} - \Delta H_{\text{soln}}$), never a stored duplicate.

Thus component resolution is **case-local overlay > standard > local (private, announced)**. This is an evidence ladder, not merely a search path: moving a file up the ladder requires review and validation, not a rename.

2.5 The ChemSep import route

The ChemSep 8.3 importer (`bin/curate/chemsep_to_choupo.py`) is deliberately reproducible and reviewable. It runs CAS-based de-duplication against the standard catalogue, then a structural/provenance audit and an internal $P^{\text{sat}}(T_b)$ closure check on every imported component record — a closure check on the imported constants and predictive correlation, not an independent comparison with laboratory data.

The same import stages NRTL, UNIQUAC and Wilson binary-pair records, each passing identity, direction, licence, source-digest and exact unit-conversion checks. Their source does not provide a captured temperature-validity range, so they remain screening data and the runtime prints `[local] UNVERIFIED` when one is selected. Pair resolution is **case-local/inline > standard > local > announced ideal default**; local data can therefore extend a case without silently changing an approved standard.

The source boundary is intentional. The unmodified third-party database and its Artistic-2.0 licence are kept under `thirdParty/chemsep/` (gitignored); the normalised, attributed records are written to the *private data/local/* tier — they are never redistributed with the public repository. Literature values retain literature provenance; Ambrose–Walton and other closures retain predictive provenance. Missing C_p , formation, electrolyte, adsorption, kinetic or transport data are not inferred merely to make a record look complete.

The case additionally owns its *equipment physics*: reaction kinetics in `constant/reactions` (the heat of reaction always computed from the formation datum and announced), per-machine measured curves (a dryer's falling-rate data) in the unit's own folder, all inherited through the LOUD fractal cascade (plant → sector → unit).

2.6 The reference case

The whole architecture is exercised by one designed flagship, `tutorials/plant/lithiumBrinePlant` (under construction): salar brine through Pitzer ponds and crystallisers, NF membrane Mg/Li separation, solvent extraction with a student-defined extractant (the γ - φ world and the audited Pitzer↔NRTL seam), electrodialysis, 60-bar CO₂ carbonation (the φ - φ override and the Henry world), a radical-kinetics burner satellite for thermal NO_x, batch recrystallisation recipes, a control satellite, and the full outer ring (fitting, sweeps, optimisation, sizing, costing, economics). Later chapters refer to its sectors by name.

2.7 The principle

No silent crutch, anywhere: every value is cited, estimated-and-labelled, overlaid-and-announced, or asserted-with-a-why — and the run log is where the whole assembly confesses. There is no ultimate model; hunting with the cat is honest engineering, hiding the cat is fraud.

3 The thermodynamic chain: how every equation here is derived

Every K-value, activity coefficient and Henry constant in this guide falls out of ONE chain of ideas. This section walks it once, so that no equation below ever has to be taken on faith. (Full derivations: the Theory Guide, chapters on flash, activity models, electrolytes and Henry's law.)

3.1 1. Equilibrium is a Gibbs minimum

The second law says an isolated system maximises entropy; recast at the conditions a process engineer actually controls — fixed T and P — it says a system minimises its **Gibbs energy** G . For a multicomponent, multiphase system,

$$dG = -S dT + V dP + \sum_i \mu_i dn_i, \quad \mu_i \equiv \left(\frac{\partial G}{\partial n_i} \right)_{T, P, n_{j \neq i}},$$

and at the minimum, moving a mole of i between phases must change nothing:

$$\boxed{\mu_i^\alpha = \mu_i^\beta} \quad \text{for every component, across every pair of phases.}$$

That single boxed statement is ALL the physics; everything else is bookkeeping to make it computable.

3.2 2. Fugacity: the workable currency

μ_i is awkward: its absolute value is unknowable and it dives to $-\infty$ as $x_i \rightarrow 0$. Lewis (1901) defined the **fugacity** f_i through $\mu_i = \mu_i^\circ + RT \ln(f_i/f_i^\circ)$ — an “effective pressure” that equals $y_i P$ in an ideal gas. Because $\ln$ is monotonic, the equilibrium condition becomes

$$\boxed{f_i^\alpha = f_i^\beta}$$

— equal fugacities, phase by phase, component by component. Every VLE “world” in Choupo is just a different, self-consistent way of WRITING f_i for a phase.

3.3 3. Two routes to a mixture fugacity

Route A — an equation of state (works for any phase the EoS can represent). Define the fugacity coefficient $\varphi_i \equiv f_i/(y_i P)$; thermodynamics gives it exactly from the volumetric behaviour,

$$RT \ln \varphi_i = \int_V^\infty \left[\left(\frac{\partial P}{\partial n_i} \right)_{T, V, n_j} - \frac{RT}{V} \right] dV - RT \ln Z,$$

so ONE model of $P(T, V, \mathbf{n})$ — a cubic like SRK with its $a(T)$, b and the pair correction k_{ij} — yields φ_i for BOTH the vapour root and the liquid root of the same cubic.

Route B — an activity model (liquids only). Split the liquid fugacity into a reference times a correction, $f_i^L = \gamma_i x_i f_i^\circ$, where the **activity coefficient** γ_i comes from a model of the excess Gibbs energy (G^E : NRTL, UNIQUAC, Wilson, Pitzer, eNRTL) via $RT \ln \gamma_i = \partial G^E / \partial n_i$. The price of Route B is choosing the **reference** f_i° — and that choice is a CONVENTION, not physics:

Convention	Reference f_i°	Normalised so
symmetric (Raoult)	pure liquid: $P_i^{\text{sat}} (\times \varphi_i^{\text{sat}} \cdot \text{Poynting at high } P)$	$\gamma_i \rightarrow 1$ as $x_i \rightarrow 1$
unsymmetric (Henry)	infinite dilution: $H_{i, \text{soln}} = \lim_{x \rightarrow 0} f_i/x_i$	$\gamma_i^* \rightarrow 1$ as $x_i \rightarrow 0$
electrolyte (aqueous)	ion at infinite dilution, $H^+(\text{aq}) = 0$ datum	$\gamma_{\pm} \rightarrow 1$ as $m \rightarrow 0$

A supercritical solute HAS no pure liquid, so its only definable anchor is the Henry limit — which is why dissolved H_2/N_2 take the unsymmetric convention, never an Antoine curve extrapolated above T_c .

3.4 4. One Gibbs surface per phase — the legality rule

Gibbs–Duhem, $\sum_i x_i d \ln \gamma_i = 0$ at constant T, P , ties every component's γ in a phase to every other's: they must all derive from ONE G^E (or one EoS) surface. Hence the rule that governs Choupo's **propertyPackage** grammar: *models per phase; conventions per component-group within the phase; correlations per component; parameters per pair*. Mixing two γ -models (or two cubics) in one phase violates Gibbs–Duhem and is refused. Mixing CONVENTIONS in one phase is legal — solvent symmetric, solutes unsymmetric — because a convention only renormalises the same surface.

3.5 5. The four worlds' K-values, assembled

Setting $f_i^L = f_i^V$ with each route/convention gives every K-value this guide uses:

world	f_i^L	f_i^V	$K_i = y_i/x_i$
γ - φ	$\gamma_i x_i P_i^{\text{sat}}$	$y_i \varphi_i P$	$\frac{\gamma_i P_i^{\text{sat}}}{\varphi_i P}$
Henry (KK/KI)	$\gamma_i^* x_i H_i(T) e^{\bar{v}_i^\infty (P-P_s)/RT}$	$y_i \varphi_i P$	$\frac{\gamma_i^* H_i \mathcal{P}}{\varphi_i P}$
φ - φ	$x_i \varphi_i^L P$	$y_i \varphi_i^V P$	$\frac{\varphi_i^L}{\varphi_i^V}$
electrolyte	ion activities from Pitzer/eNRTL on the aqueous rung; full speciation (mass action + charge balance) where the case demands it		

Each row is one `propertyMethods.liquid` family (`activity.*`, `solution.henryDilute`, `eos.*`, `electrolyte.*`); the run log announces which world is live. All four are validated in the tutorials against measured data — van Ness-consistent VLE sets (γ - φ), Wiebe/Larson solubilities (Henry, $\sim 2\%$), the Stryjek N₂-CH₄ isotherm (φ - φ , $\sim 2\%$), and the Hamer–Wu / calorimetric electrolyte sets.

Intuition. The whole edifice is: G minimum $\rightarrow$ equal $\mu \rightarrow$ equal $f \rightarrow$ pick HOW to write f per phase (an EoS or a G^E model + a reference convention) $\rightarrow$ a K-value. When an equation in this guide looks arbitrary, walk it back up this chain — it never is.

4 Why a separate guide

The Theory Guide answers *how a model works* — how a flash splits, how an EoS predicts Z , how Levenberg–Marquardt minimises a residual. This guide answers a question that comes *before* that: *where do the numbers that feed the models come from?* A simulation is only as good as the T_c , the NRTL pair, the heat of formation it stands on, and those are not handed down — they are estimated, fitted, and tested.

Choupo treats that as a first-class phase with its own binary (`choupoProps`) and its own lifecycle:

CREATE a component (from groups) → **CURATE** the mixture models (fit pairs, test data, choose by evidence) → **WIRE** the flowsheet → **SIMULATE**.

A `propsDict` of `operations` drives the first two stages. A case may have *no* `flowsheetDict` at all (a pure properties case — the compound bench), or carry one and use props to curate the thermo it will simulate. Same files, same engine; only the centre of the GUI differs.

Intuition. The usual reflex is to click a property method and trust a badge. Choupo’s stance (the “see, then decide” principle) is the opposite: you make the estimate or the fit *visible* — the group build-up, the model curve against the lab points, the consistency residual — and decide from the evidence in front of you. Estimation has error; the point is to see it.

5 Where curated data lives, and how a case selects it

You estimated a T_c , fitted an NRTL pair, tested a heat of solution. Two questions follow: *where does that number land*, and *how does a case use it*? Choupo’s answer is *declarative property packages with OpenFOAM-like explicit files*.

A curated value lands in a glass-box `.dat` in a tree organised by *what kind of thing it is* — the intrinsic property of one substance, the parameter of a pair, the equilibrium of a reaction:

```
data/standards/
  components/<name>.dat      intrinsic, one substance (Tc, Antoine, Cp)
                             + dissociatesTo { Na 1; Cl 1; } [salts]
  species/aqueous.dat        one catalogue; each record a model species
                             (Na, Cl, CO2aq): charge + medium-tagged
                             thermo; never fed to a flowsheet
  phases/solid/<phase>.dat    crystal phase (rho_p, k_v): halite, sylvite
  chemistry/.../<name>.dat    an equilibrium: dissolution, speciation
  parameters/.../<i>-<j>.dat   a pair: NRTL, Pitzer, eNRTL
  propertyMethods/.../<name>.dat a model declaration (+ its reference basis)
  (the package is the manifest INLINE in the case's constant/propertyDict --
   THE CENTRE; there is no shared propertyPackages/ catalogue)
```

The arity rule decides *which* file: a number that “stands alone with the molecule” (the T_c of water) lives with the component; a number that needs two names (an NRTL pair) cannot, and goes to `parameters/`. The Developer Guide (*the thermophysical property architecture*) gives the full layout and the eight data kinds.

Selecting a package. A case does not list loose models and parameters. It *selects* a `propertyPackage` — one record that names the components, the methods, and the parameter banks the run needs:

```
recordType propertyPackage;  name aqueousNaCl_pitzer;
components      ( water NaCl ); // NaCl's dissociatesTo names the ions
chemistry       { salts ( halite ); } // the builder reads salts; the
                                     // solubility anchor comes from the
                                     // salt record itself
propertyMethods { liquid electrolyte.pitzer; vapour builtin.idealGas; }
parameters      { pitzerPairs { Na-Cl ".../pitzer/pairs/Na-Cl.dat"; } }
% NOTE: pitzerPairs is today a CITATION of where the pair lives; the loader
% resolves the pair by (cation,anion) name. kijPairs/henryPairs/binaryPairs
% are loaded from their declared paths and refuse when missing.
```

The flow is one sentence: **propertyPackage declares, ThermoPackageBuilder assembles, ThermoPackage computes**. The package is the recipe; the builder loads what it names and mounts the runtime object the unit operations consume. An ordinary molecular case is the same record degenerate — no dissociation, no chemistry, an NRTL or EoS model, optional binary pairs; the old flat `thermoPackage` dict you already write *is* that degenerate package.

Intuition. The boundary you must respect: *curation estimates, runtime does not*. You may estimate, fit, and derive at the props bench — and you *see* it (§6, §7). Once a case runs, the builder only *loads* existing case, standard or local records; it never invents a missing property. For activity models, an unresolved binary pair may use the documented ideal default, but that fallback is announced and appears in the provenance. A required record with no defined fallback stops the run with a clear message. So: estimate, look, promote — and inspect every local or defaulted rung used by the package.

6 Creating a component: Joback group contribution

When a component is missing from the catalogue and from the literature, you can *estimate* its pure-component constants from its molecular structure — the groups it is built from. Choupo implements the **Joback** method (Joback & Reid, 1987; tabulated in Poling, Prausnitz & O’Connell, *The Properties of Gases and Liquids*, 5th ed.), the `estimateComponent` property operation.

6.1 The method

Each first-order group contributes a tabulated increment. Summing the increments and substituting into Joback’s correlations gives the constants (with n_A the total number of atoms):

$$T_b = 198.2 + \sum_i n_i \Delta T_{b,i} \quad (4)$$

$$T_c = T_b / (0.584 + 0.965 \Sigma \Delta T_c - (\Sigma \Delta T_c)^2) \quad (5)$$

$$P_c = (0.113 + 0.0032 n_A - \Sigma \Delta P_c)^{-2} \text{ [bar]} \quad (6)$$

$$V_c = 17.5 + \sum_i n_i \Delta V_{c,i} \text{ [cm}^3\text{/mol]} \quad (7)$$

$$\Delta H_f^\circ(298) = 68.29 + \sum_i n_i \Delta H_{f,i} \text{ [kJ/mol, ideal gas]} \quad (8)$$

$$\Delta H_{\text{vap}}(T_b) = 15.30 + \sum_i n_i \Delta H_{v,i} \quad (9)$$

$$C_p^{\text{ig}}(T) = (\Sigma a - 37.93) + (\Sigma b + 0.210)T + (\Sigma c - 3.91 \times 10^{-4})T^2 + (\Sigma d + 2.06 \times 10^{-7})T^3. \quad (10)$$

The acentric factor ω is *not* a group sum; Choupo derives it from (T_b, T_c, P_c) by the Lee–Kesler correlation, with $T_{br} = T_b/T_c$ and P_c in atm:

$$\omega = \frac{-\ln P_c - 5.97214 + 6.09648/T_{br} + 1.28862 \ln T_{br} - 0.169347 T_{br}^6}{15.2518 - 15.6875/T_{br} - 13.4721 \ln T_{br} + 0.43577 T_{br}^6}. \quad (11)$$

A few first-order group increments (Joback & Reid; the full table is in the code, `src/propertyOps/EstimateComponent.cpp`):

group	ΔT_b	ΔT_c	ΔP_c	ΔH_f (kJ/mol)
–CH ₃	23.58	0.0141	–0.0012	–76.45
>CH ₂	22.88	0.0189	0.0000	–20.64
–OH (alcohol)	92.88	0.0741	0.0112	–208.04
>C=O (ketone)	76.75	0.0380	0.0031	–133.22
aromatic =CH–	26.73	0.0082	0.0011	2.09

6.2 The structure is identity: jobackGroups and bin/estimate

The molecular structure belongs *with the substance*, like the formula: a component’s `.dat` declares its groups once,

```
jobackGroups { CH3 2; ketone 1; }      // acetone = 2 x CH3 + 1 x >C=O
```

(the sibling of the UNIFAC `groups{}` block — a different table, the same idea), and estimation needs no case at all:

```
bin/estimate acetone      # case-local first, then the standard catalogue
bin/estimate path/to/new.dat  # or an explicit file
```

The command runs the Joback engine on the component’s own groups, prints the full glass-box build-up (each group’s increment, the closure chain, deviations against any **reference** block), and drops *one* stable proposal — `<name>.estimated.dat` — next to the source, for you to review and promote by hand. It never writes into the frozen `data/standards/` tree, and the solver still refuses to estimate at run time: estimation is a *curation* act with a paper trail, not a silent fill-in.

The `estimateComponent` operation remains the pedagogical form — declare the groups inline in a `propsDict` when you want the estimate itself to be the lesson, as in the worked example below.

6.3 Worked example: acetone

Acetone, $\text{CH}_3-\text{CO}-\text{CH}_3$, decomposes into $2 \times \text{CH}_3$ and $1 \times \text{ketone}$. The operation prints the build-up and validates against a **reference** block:

Property	Joback	Reference (NIST)	Dev.
T_b	322.1 K	329.2 K	-2.2%
T_c	500.6 K	508.1 K	-1.5%
P_c	48.0 bar	47.0 bar	+2.2%
V_c	209.5 cm ³ /mol	209 cm ³ /mol	≈ 0
ω	0.295	0.307	-3.9%
ΔH_f°	-217.8 kJ/mol	-217.1 kJ/mol	+0.3%

Within a few percent on the criticals and almost exact on ΔH_f — and you *see* the error, because the reference sits beside the estimate.

Intuition. Know the method's limits. Joback is weak on T_b of strongly hydrogen-bonding species (ethanol's T_b comes out ~ 14 K low), and it yields *no* Antoine / vapour-pressure parameters — those need a fit (§7) or a corresponding-states correlation, a separate step. An estimate is a starting point you then anchor to data, never a final answer.

Runnable case. `tutorials/props/estimate/estimate_acetone` — the example above, groups in, validated estimate out.

7 Curating mixture models: fitting binary pairs

A pure-component set is not enough for a mixture: the non-ideality lives in the *binary interactions* (NRTL, Wilson). Those parameters are regressed against experimental VLE data by Levenberg–Marquardt — the `fitParameters` operation.

7.1 The activity-coefficient models

For a binary, the **NRTL** model gives the liquid-phase activity coefficients from three adjustable numbers per pair — τ_{12}, τ_{21} (temperature-dependent) and the non-randomness α :

$$\tau_{ij} = a_{ij} + \frac{b_{ij}}{T}, \quad G_{ij} = \exp(-\alpha \tau_{ij}), \quad (12)$$

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{\tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2} \right], \quad (13)$$

and $\ln \gamma_2$ by swapping the indices. So a pair carries four fittable coefficients ($a_{12}, b_{12}, a_{21}, b_{21}$) plus a fixed α (often 0.3). **Wilson** is the two-parameter alternative,

$$\ln \gamma_1 = -\ln(x_1 + \Lambda_{12}x_2) + x_2 \left[\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1} \right], \quad (14)$$

which cannot represent a liquid–liquid split (use NRTL when one is suspected).

7.2 The residual the fit minimises

The measurable is the bubble temperature. At a fixed pressure P and liquid composition x , the bubble point is the T that closes modified Raoult,

$$\sum_i x_i \gamma_i(x, T) P_i^{\text{sat}}(T) = P \quad (15)$$

(a Newton-1D in T ; P_i^{sat} from Antoine). Levenberg–Marquardt then drives the pair coefficients $\theta = (a_{12}, b_{12}, a_{21}, b_{21})$ to minimise the sum of squared bubble-temperature residuals against the data,

$$\min_{\theta} \sum_k \left[T_{\text{bub}}(x_k; \theta) - T_k^{\text{exp}} \right]^2. \quad (16)$$

Intuition. A low residual is not a good fit. At a *single* pressure $\tau_{ij} = a_{ij} + b_{ij}/T$ is nearly linear in both a_{ij} and b_{ij}/T over the narrow T range of one isotherm/isobar, so a and b are *correlated* — many (a, b) pairs fit equally well and the optimum is a valley, not a point. `fitParameters` forms $J^T J$ at the optimum and reports the parameter correlation + confidence intervals so you SEE this; the Theory Guide’s *Levenberg–Marquardt* and *What is fittable* chapters derive the diagnostic. Fit across pressures, or fix b , to break it.

7.3 See, then decide: the model overlay

Before fitting, look. Declare one scan `operation` per candidate model (ideal, NRTL, Wilson), emitting the $(x, T_{\text{bubble}}, y_{\text{eq}})$ trio, and an `experimental` `{}` entry that overlays them — and, with a `dataset`, draws the measured points on top. The student sees which model the data actually picks: for ethanol/water at 1 atm the ideal curve misses the azeotrope entirely while NRTL lands on it.

Runnable case. `tutorials/props/compare/compare_vle_etoh_water` — ideal vs. NRTL bubble/dew curves overlaid with measured VLE points (Carey & Lewis, 1932).

7.4 Fit, then promote

When a model earns its keep, `fitParameters` regresses its coefficients to the data and writes a proposal to `constant/binaryPairs/<model>/<pair>.fit-<date>.dat`. Promotion is a deliberate, visible act: you inspect the parity plot and the identifiability statistics (confidence intervals, parameter correlation) the fit reports — never a “good fit” badge — and copy the proposal into the case’s `constant/` when you trust it.

8 Testing the data itself: consistency

Before fitting a model *to* data, test whether the data is even thermodynamically consistent — a fit to bad data is a polished wrong answer. The `vleConsistency` operation does this in three steps.

8.1 Step 1 — activity coefficients straight from the data

It does not assume a model. From each measured (x_1, y_1, T) at the run pressure P , modified Raoult gives the activity coefficient of each component directly:

$$\gamma_i = \frac{y_i P}{x_i P_i^{\text{sat}}(T)}, \quad P_i^{\text{sat}}(T) \text{ from Antoine.} \quad (17)$$

These are *experimental* γ_i — the test now asks whether they obey thermodynamics.

8.2 Step 2 — the Gibbs–Duhem point test

The Gibbs–Duhem equation is the constraint every real mixture must satisfy at constant T, P :

$$\sum_i x_i d \ln \gamma_i = 0 \implies x_1 \frac{d \ln \gamma_1}{dx_1} + x_2 \frac{d \ln \gamma_2}{dx_1} = 0 \quad (\text{binary}). \quad (18)$$

The operation evaluates the left-hand side at each interior point (central differences in x_1) and reports its max and mean magnitude — the *point residual*. A consistent set keeps it near zero. (Isobaric data adds a small $-H^E/RT^2 \cdot dT/dx$ term, dropped here; stated as a caveat, not hidden.)

8.3 Step 3 — the Herington area test

Integrate (18) across the whole composition range: for an isothermal set Gibbs–Duhem forces the net area of $\ln(\gamma_1/\gamma_2)$ to vanish. Herington’s isobaric form compares that net area to the absolute area and corrects for the boiling-range:

$$D = 100 \frac{\left| \int_0^1 \ln \frac{\gamma_1}{\gamma_2} dx_1 \right|}{\int_0^1 \left| \ln \frac{\gamma_1}{\gamma_2} \right| dx_1}, \quad J = 150 \frac{T_{\text{max}} - T_{\text{min}}}{T_{\text{min}}}, \quad (19)$$

(both integrals by the trapezoid rule over the sorted points). The set **passes** when $|D - J| < 10$. The GUI’s consistency card plots $\ln \gamma_1$, $\ln \gamma_2$ and $\ln(\gamma_1/\gamma_2)$ vs x_1 with the net area shaded — you SEE the $\pm$ areas cancel, not just the number.

Finally the infinite-dilution coefficients $\gamma_1^\infty, \gamma_2^\infty$ (the dilute-end limits) are reported — the most stringent point of any model later fitted to the set.

Runnable case. `tutorials/props/compare/compare_vle_etoh_water` — the `vleConsistency` op on the Carey–Lewis ethanol/water data (Herington $|D - J| = 4.6 < 10$, PASS).

This guide grows with the props pillar. Planned: a second group-contribution method (Constantinou–Gani) to compare against Joback; group contribution for liquid viscosity; a structure (SMILES) front-end for the compound bench.